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Microstructural Brain Reserve Moderates the Associations Between 24-h Movement Behaviors and Cognition in Older Adults: Cross-Sectional Findings From the AGUEDA Trial

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ABSTRACT

This study examined the associations between 24-h movement behaviors and cognitive function in cognitively unimpaired older adults, and whether Alzheimer's disease brain signatures, including gray matter mean diffusivity (GMMD) and thickness/volume signatures, moderated these associations. A total of 91 participants were enrolled in the AGUEDA trial (NCT05186090), with 89 included in this cross-sectional analysis (mean age = 71.61 ± 3.85 years; 56.2% females). Moderate-to-vigorous physical activity (MVPA), light physical activity (LPA), sedentary behavior (SB), and sleep were objectively assessed using 9-day wrist-worn accelerometry. Cognitive function was assessed across attention/inhibitory control, episodic memory, processing speed, visuospatial processing, and working memory. Thickness/volume and GMMD signatures were derived from cortical and hippocampal regions. Compositional linear regression models (CoDA) examined associations, adjusting for age, sex, and education, and tested interactions between movement behaviors and Alzheimer's disease brain signatures. No direct associations were observed between 24-h movement behaviors and cognitive domains (all $p > 0.05$). However, GMMD signature, but not thickness/volume signature, moderated several movement behavior-cognition associations. Moderation was most evident for episodic memory, with significant interactions for MVPA, LPA, and sleep, while LPA exhibited the broadest pattern of moderation across cognitive domains, interacting with attention/inhibitory control, episodic memory, processing speed, and working memory (all p for interaction ≤ 0.04). In conclusion, the associations between 24-h movement behaviors and cognitive function may depend on microstructural brain integrity (i.e., GMMD). These findings highlight the importance of considering Alzheimer's disease-related brain signatures when investigating lifestyle–cognition associations and support a more personalized approach to promoting cognitive health in older adults. Future prospective longitudinal studies and CoDA interventional

Abbreviations: AD, Alzheimer's disease; CoDA, Compositional Data Analysis; DCCST, Dimensional Change Card Sort Test; DSST, Digit Symbol Substitution Test; GMMD, gray matter mean diffusivity; ILR, isometric log-ratio; LPA, light PA; MMSE, mini-mental state examination; MoCA, montreal cognitive assessment; MVPA, moderate-to-vigorous PA; PA, physical activity; PSMT, Picture Sequence Memory Test; ROFT, Rey–Osterrieth complex figure test; SB, sedentary behavior; STICS-m, Spanish telephone interview for cognitive status; SWMT, Spatial Working Memory Test; TMT, Trail Making Tests.

trials are needed to establish temporal precedence and evaluate whether targeted time reallocations can actively attenuate cognitive trajectories across distinct microstructural vulnerability profiles.
Trial Registration: NCT05186090

1 | Introduction

Aging is accompanied by progressive changes in brain structure and function that increase vulnerability to cognitive decline and Alzheimer's disease (AD) [1]. Even in cognitively normal older adults, subtle alterations in brain reserve, particularly in AD-related cortical regions and microstructural tissue properties, may precede measurable cognitive impairment [2]. Brain reserve refers to the structural and biological resources of the brain (i.e., such as the number of neurons, synapses, and overall brain structure), which may help some individuals better tolerate age- or disease-related brain changes before clinical symptoms emerge [3]. Identifying modifiable lifestyle factors that slow down these early markers of brain vulnerability is therefore critical for preserving cognitive function during aging [4].

Accumulating evidence suggests that 24-h movement behaviors, including physical activity (PA), sedentary behavior (SB), and sleep, play an important role in brain and cognitive health [5–12]. Higher levels of PA are generally associated with better cognitive performance [13–16] and greater total and medial temporal gray matter volumes [17]; whereas prolonged sedentary time [18] has been associated with lower total brain volume [11], and inadequate sleep duration (< 7 h and more than 8 h a day, respectively) with poorer cognitive outcomes [19]. However, findings across studies remain inconsistent, with several investigations reporting weak or null associations, particularly in cognitively unimpaired populations [9, 20, 21]. One potential explanation for these inconsistencies is the assumption that the cognitive effects of 24-h movement behaviors are uniform across individuals, without considering underlying differences in brain morphology and physiology [22–24].

Recent advances in neuroimaging have enabled the characterization of AD-related brain signatures in cognitively unimpaired individuals [25–27]. These signatures were selected because they capture structural features of regions known to be vulnerable to AD, including both cortical thickness and the volume of AD-related regions such as the hippocampus [28, 29]. In addition, gray matter mean diffusivity (GMMD) has emerged as a sensitive marker of microstructural brain disruption, reflecting early tissue integrity beyond macroscopic atrophy [25]. Importantly, GMMD alterations have been linked to early neurodegenerative and cerebrovascular processes and may identify individuals at higher risk for cognitive decline before clinical symptoms emerge [30, 31].

From a precision medicine perspective, it is plausible that the associations between 24-h movement behaviors and cognitive function depend on the underlying state of brain reserve [3, 32]. Rather than assuming a uniform response, neurobiological frameworks suggest a dynamic and reciprocal relationship between movement patterns and brain reserve [22], where a brain's capacity to benefit from behavioral stimuli is conditioned by its underlying tissue state. Specifically, individuals with higher microstructural brain reserve may benefit more from balanced movement patterns across the day, whereas those with greater microstructural disruption may require higher-intensity physical activity to support cognitive performance [33]. However, to our knowledge, this is one of the first studies to examine whether AD-related brain signature scores modify the relationships between 24-h movement behaviors and specific cognitive domains in cognitively unimpaired older adults using compositional data analysis (CoDA) approach, which appropriately accounts for the co-dependency of time-use behaviors [34].

Therefore, the present study aimed to examine the associations between 24-h movement behaviors and multiple cognitive domains related to aging, including attentional/inhibitory control, episodic memory, processing speed, visuospatial processing, and working memory, in cognitively unimpaired older adults, and to investigate whether these associations differed according to levels of AD brain signature scores. By integrating compositional analyses, moderation, subgroup, and time-reallocation approaches, this study seeks to provide a more nuanced understanding of how daily movement behaviors relate to cognition in the context of early brain vulnerability to AD markers. We hypothesized that the associations between 24-h movement behaviors and cognitive function would be moderated by microstructural brain reserve, such that individuals with greater microstructural vulnerability (higher GMMD) would show more pronounced associations with higher-intensity physical activity.

2 | Material and Methods

2.1 | Design and Participants

This study used baseline data from the AGUEDA (Active Gains in brain Using Exercise During Aging) trial. The AGUEDA trial was a single-center, two-arm, single-blind randomized controlled trial that randomized 91 cognitively unimpaired older adults (65–80 years old) from Granada (Spain). All data were collected between March 2021 and May 2022. The trial protocol followed the principles of the Declaration of Helsinki and was approved by the Research Ethics Board of the Andalusian Health Service (CEIM/CEI Provincial de Granada; #2317-N-19) on May 25, 2020. The trial was registered on ClinicalTrials.gov (Identifier: NCT05186090; Submission date: December 22, 2021). All participants provided written informed consent. Details of the methodology and protocols of the AGUEDA project have been described elsewhere [35] and deposited at GitHub (https://github.com/aguedaprojectugr).

2.2 | Eligibility Criteria

Eligibility criteria were defined as: (i) older adults between 65 and 80 years old, (ii) physically inactive (i.e., defined as not participating in any resistance exercise (RE) program in the last 6 months or accumulating < 600 METs-min/week per the International Physical Activity Questionnaire (IPAQ)) [36], (iii) cognitively unimpaired (≥ 26 points) according to the Spanish version of the modified Telephone Interview of Cognitive Status (STICS-m) [37], Mini-Mental State Examination (MMSE) (≥ 25/30) [38] and Montreal Cognitive Assessment (MoCA) (< 71 years: ≥ 24/30; 71–75 years: 22/30; > 75 years: 21/30) [39]; and (iv) non-significant depressive symptoms at baseline according to the Geriatric Depression Scale (GDS) (score < 15) [40]. Detailed information about eligibility criteria is available elsewhere [35].

2.3 | Outcome Measurements

2.3.1 | 24-h Movement Behaviors

Moderate-to-vigorous PA (MVPA), light intensity PA (LPA), SB, and sleep were measured using a wrist-worn Actigraph GT3X+ accelerometer device (Pensacola, FL, USA) for nine consecutive 24-h periods (except for water activities) following the same approach as our previous work [10]. To assist the sleep-detection algorithm, participants were instructed to record their bedtime and wake-up time in a sleep diary. The accelerometer was set to capture raw acceleration data at a sampling rate of 100 Hz. These raw signals were calibrated [41] and used to detect non-wear periods based on the variability and magnitude of the data [42]. The signals were then converted into 5-s epochs, computing the Euclidean Norm of the acceleration with negative values set to zero. Specific thresholds were applied to classify activity intensity: SB was defined as < 35 mg, LPA as 35–100 mg, and MVPA as > 100 mg [43, 44]. To exclude sporadic arm movements from the MVPA classification, bouts of at least 1 consecutive minute with a gap tolerance of 80% were considered an indicator of MVPA (≥ 48 s/min), following previous reports [45]. Data were included in the analysis only if participants had at least four valid days (with a minimum of 16 h of wear time per day), including one weekend day. Daily averages for sleep, SB, LPA, and MVPA were calculated using valid days.

2.3.2 | Cognitive Domains

Participants underwent a comprehensive neuropsychological evaluation assessing five cognitive domains. Cognitive domains were established by replicating the approach used in a large-scale RCT with similar participant characteristics, cognitive tests, and aims, but with a considerably larger sample size ($n = 648$) that enabled multiple-factor Confirmatory Factor Analysis (CFA) [46]. Inverse variables were recoded when necessary, so that higher scores in all cognitive-domain composites indicate better performance. For each cognitive test, z-scores were calculated by subtracting individual raw values from the group mean and dividing by the group standard deviation. Cognitive-domain scores were computed as the average of the standardized z-score from the individual tests. Missing cognitive test scores were not imputed. Cognitive domain composite scores were calculated when participants had valid scores for all tests or were missing only one cognitive test within that domain. Consequently, no missing data were present for any final cognitive domain composite score at baseline. The five cognitive domains assessed included: (i) attentional/inhibitory control (Trail Making Test [Time part B] [47], Dimensional Change Card Sort [Computed score] [48], Stroop test [Incongruent raw score], Flanker test [Computed score] [48]); (ii) episodic memory (MoCA delayed recall [Multiple choice cue] [49], Picture Sequence Memory Test [Raw score] [48], Rey Auditory Verbal Learning Test [Total recall, delayed and recognition raw score] [50], Rey—Osterrieth Complex Figure Test [Copy and delayed raw score] [51]); (iii) processing speed (Digit Symbol Substitution Test [Total correct raw score] [52] and Trail Making Test [Time part A] [47]); (iv) visuospatial processing (Wechsler Adult Intelligence Scale with Matrix Reasoning and Block Design [Accuracy] [53] and MoCA Clock Draw [Total Score] [49]); and (v) working memory (Spatial Working Memory Test [3 and 4 item] [54]), List Sorting Working Memory Test [Total correct] [55], N-Back Test [2-back] [56].

Detailed information about cognitive domains is available elsewhere [57].

2.3.3 | AD Brain Signatures

GMMD and thickness/volume signatures were derived according to the methodology described by Williams et al. [25] using a data-driven approach to identify brain regions sensitive to early AD-related neurodegeneration [58]. The thickness/volume signature consisted of a weighted composite of cortical thickness from seven bilateral regions (ROIs: entorhinal cortex, middle temporal gyrus, banks of the superior temporal sulcus, superior temporal gyrus, isthmus cingulate, lateral orbitofrontal cortex, and medial orbitofrontal cortex) together with hippocampal volume.

Structural and diffusion MRI measures from each ROI were adjusted for age, while hippocampal volume was additionally adjusted for estimated intracranial volume. Standardized residuals were computed for each ROI and subsequently weighted and summed to derive the thickness/volume signature scores. The same weights used for structural data were applied to the GMMD values of each ROI, including the hippocampus, to compute the GMMD signature score. Higher GMMD signature values reflect poorer brain reserve [59], while higher thickness/volume signature are indicative of greater cortical reserve [60]. Both AD brain signature scores were expressed as z-scores using the baseline sample mean and standard deviation. More information regarding the preprocessing and processing of MRI data is reported in a previous study [61].

2.3.4 | Moderation Outcomes and Covariates

AD brain signatures (i.e., GMMD and thickness/volume signatures) were examined as continuous moderators in interaction models evaluating whether associations between 24-h movement behavior compositions and cognitive outcomes differed according to levels of AD-related brain vulnerability. Additional subgroup analyses were conducted after stratifying participants according to median levels of GMMD and thickness/volume signatures (lower than median vs. higher than median groups). Sex, age, and years of education were included as covariates in cognitive analyses. Analyses including AD brain signatures were adjusted for sex and education, as they had already been previously corrected for age.

2.4 | Statistical Analyses

2.4.1 | Associations of 24-h Movement Behaviors With Cognition

Analyses were conducted in R 4.5.2 with the package *GGIR* (version 3.1-1) [45], *compositions* [62] and *robCompositions* [63]. The characteristics of the study sample were calculated using means and standard deviations (SD) for continuous variables and frequencies and percentages for categorical variables. Pearson correlation analyses were performed to examine bivariate associations between cognitive outcomes and AD brain signatures.

Movement behaviors were treated as compositional data and defined as the daily time (minutes/day) accumulated in MVPA, LPA, SB, and sleep. First, the composition was closed to 1440 min and after checking for the presence of zero values in our behavioral variables, no zeroes were observed in any of the components and, therefore, imputation was not needed. Next, four sets of three isometric log ratios (ILR) were calculated, each set rotated the dominant behavior for the composition (i.e., MVPA, LPA, SB, or sleep), and the coefficients for the first ILR in the set were reported [64, 65]. Subsequently, the ILR were fit as predictors along with the covariates to investigate the association of movement behaviors with the outcomes in linear regression models. Detailed information about the ILR calculations is available elsewhere [10].

Linear regression models were fitted following CoDA approaches to examine the associations between the daily time accumulated in movement behaviors (i.e., MVPA, PA, SB and sleep, expressed as ILR) and cognitive domains. The strength and direction of the relationship between each behavior in relation to all remaining behaviors (such as MVPA compared to the geometric mean of LPA, SB and sleep) and an outcome (for instance, episodic memory) are indicated by the model coefficients (unstandardized coefficient). Point estimates are presented with 95% confidence intervals to emphasize the magnitude and uncertainty of the associations. Exploratory subgroup and reallocation results were interpreted by considering both the direction and precision of estimates, rather than focusing exclusively on a preset alpha threshold.

2.4.2 | Moderation, Subgroup, and Exploratory Analyses

Moderation by AD brain signatures was first assessed using ANOVA F-tests comparing nested linear regression models with and without the three ILR × moderator interaction terms. This approach provided a single F statistic and p-value for the interaction between the overall 24-h movement behavior composition and each AD brain signature for each cognitive outcome. GMMD and thickness/volume signatures were tested in separate models.

Second, behavior-specific pivot ILR interaction models were fitted to examine whether the association between each movement behavior contrast and cognitive outcomes differed according to AD brain signature levels. These models included interaction terms between the first pivot ILR coordinate and continuous AD brain signature z-scores. Therefore, the interaction coefficients indicated whether the association between a given behavior, expressed relative to the remaining behaviors, and cognitive outcomes varied according to GMMD or thickness/volume signature levels. Additional subgroup analyses stratified by median levels of AD brain signatures were performed (lower than median vs. higher than median groups) to describe the direction and magnitude of the associations.

Finally, we performed compositional isotemporal substitutions analyses (*codaredistlm* package version 0.1.0) following the significant moderation findings. We explored how meaningful reallocations of time (i.e., increasing or decreasing time in each movement behavior in 5-min increments from −5 to +30 min, using the mean time-use composition of an average participant as the reference) were associated with outcomes, using proportional (e.g., increasing time in sleep by 30 min whilst drawing time from all other behaviors pro rata) and one-for-one reallocation (e.g., increasing time in sleep by directly drawing time from MVPA). All models were checked for linearity, normality, and outliers to ensure assumptions were not violated. We set statistical significance levels at $p < 0.05$. Statistical significance in one-to-one reallocation models was inferred when the 95% confidence intervals of the regression coefficients did not cross zero. Detailed information about the CoDA approach is available elsewhere [10].

3 | Results

3.1 | Sample Characteristics

Table 1 presents the descriptive characteristics of the total sample and stratified according to lower (< median) and higher (≥ median) levels of AD brain signatures. Eighty-nine participants were included (56.2% female), with a mean age of 71.61 ± 3.85 years. Figure 1 illustrates the daily movement behavior composition using ternary plots, showing the relative distribution of time spent in MVPA, LPA, SB, and sleep, stratified by higher and lower levels of GMMD (Figure 1A) and thickness/volume signatures (Figure 1B). For the GMMD signature, participants in the higher GMMD group spent greater proportion of the day doing PA (20.42%, 294.01 min/day; 39.25 min/day of MVPA and 254.76 min/day of LPA), specifically, compared with the lower GMMD group (18.97%, 273.11 min/day; 28.32 min/day of MVPA and 244.79 min/day of LPA). In contrast, the lower GMMD group spent a slightly greater proportion of time in SB (49.05%, 706.44 min/day) than the higher GMMD group (48.55%, 699.18 min/day). When stratified by thickness/volume signature, participants in the lower thickness/volume group spent a greater proportion of time doing PA (19.82%, 285.42 min/day; 36.26 min/

TABLE 1 | Descriptive characteristics of the study sample.

	Total	Gray matter mean diffusivity signature		Thickness/volume signature	
	Mean ± SD	Lower levels	Higher levels	Lower levels	Higher levels
n	89	43[a]	42[a]	45	44
Physical characteristics					
Female (*n*, %)	50 (56.2)	22 (51.2)	27 (64.3)	26 (57.8)	24 (54.5)
Age (y)	71.61 (3.85)	71.86 (3.87)	71.24 (3.90)	71.75 (3.90)	71.47 (3.85)
Education length (y)	11.74 (4.90)	10.93 (5.17)	12.43 (4.64)	11.00 (4.09)	12.50 (5.55)
≥ 12 years education (*n*, %)	34 (38.2)	15 (34.9)	18 (42.9)	16 (35.6)	18 (40.9)
Weight (kg)	73.69 (13.41)	76.32 (14.29)	71.43 (12.52)	73.66 (13.38)	73.71 (13.60)
Height (cm)	160.62 (9.23)	163.22 (9.65)	157.84 (8.33)	160.28 (9.35)	160.96 (9.20)
Body mass index (kg/m^2)	28.49 (4.24)	28.55 (4.23)	28.63 (4.37)	28.55 (3.86)	28.43 (4.65)
≥ 12 Pathological amyloid load (*n*, %)	19 (21.3)	9 (20.9)	9 (21.4)	9 (20.0)	10 (22.7)
Apolipoprotein E4 carrier (*n*, %)	13 (14.9)	6 (14.6)	7 (16.7)	10 (22.7)	3 (7.0)
General cognition					
Spanish version of the modified Telephone Interview of Cognitive Status (score)	33.97 (2.74)	33.65 (2.77)	34.14 (2.70)	33.96 (2.39)	33.98 (3.08)
Montreal Cognitive Assessment (Raw score)	25.65 (2.09)	25.19 (2.12)	26.10 (2.03)	25.76 (2.14)	25.55 (2.05)
Mini-Mental State Examination (Raw score)	28.94 (1.06)	29.05 (1.02)	28.86 (1.12)	28.93 (1.16)	28.95 (0.96)
Cognitive domains					
Attentional/inhibitory control (z-score)	0.04 (0.97)	−0.03 (1.01)	0.06 (0.97)	0.01 (0.92)	0.06 (1.03)
Episodic memory (z-score)	0.05 (0.92)	0.04 (0.92)	0.06 (0.92)	0.08 (1.05)	0.02 (0.77)
Processing speed (z-score)	0.02 (1.00)	−0.07 (0.81)	0.06 (1.20)	−0.06 (1.18)	0.10 (0.78)
Visuospatial processing (z-score)	0.03 (0.97)	−0.17 (0.89)	0.14 (0.96)	0.04 (0.97)	0.03 (0.97)
Working memory (z-score)	0.03 (0.99)	−0.09 (1.02)	0.10 (0.95)	0.00 (1.09)	0.07 (0.88)
AD brain signatures					
Gray matter mean diffusivity signature (z-score)	−0.04 (0.97)	−0.74 (0.71)	0.67 (0.60)	−0.02 (1.02)	−0.06 (0.92)
Thickness/volume signature (z-score)	0.04 (0.96)	0.04 (0.99)	−0.05 (0.94)	−0.68 (0.58)	0.78 (0.67)
Movement behaviors					
Moderate-to-vigorous physical activity (min/day)	45.81 (33.04)	42.63 (35.79)	50.13 (31.20)	48.58 (33.59)	42.96 (32.61)
Light physical activity (min/day)	254.51 (72.49)	246.52 (75.54)	259.60 (70.39)	253.51 (68.99)	255.54 (76.68)
Sedentary behavior (min/day)	692.60 (84.29)	697.66 (85.68)	687.36 (85.18)	695.14 (82.15)	690.01 (87.30)
Sleep (min/day)	447.08 (51.39)	453.20 (48.64)	442.91 (54.26)	442.77 (48.61)	451.48 (54.29)

Abbreviations: acc, accuracy; AD, Alzheimer's disease; CL, centiloid; cm, centimeters; ilr, isometric log-ratio coordinate; m, meters; min, minutes; n, number; rt, reaction time; SD, standard deviation; sec, seconds; y, years.
[a]There are four participants with gray matter mean diffusivity signature missing data.

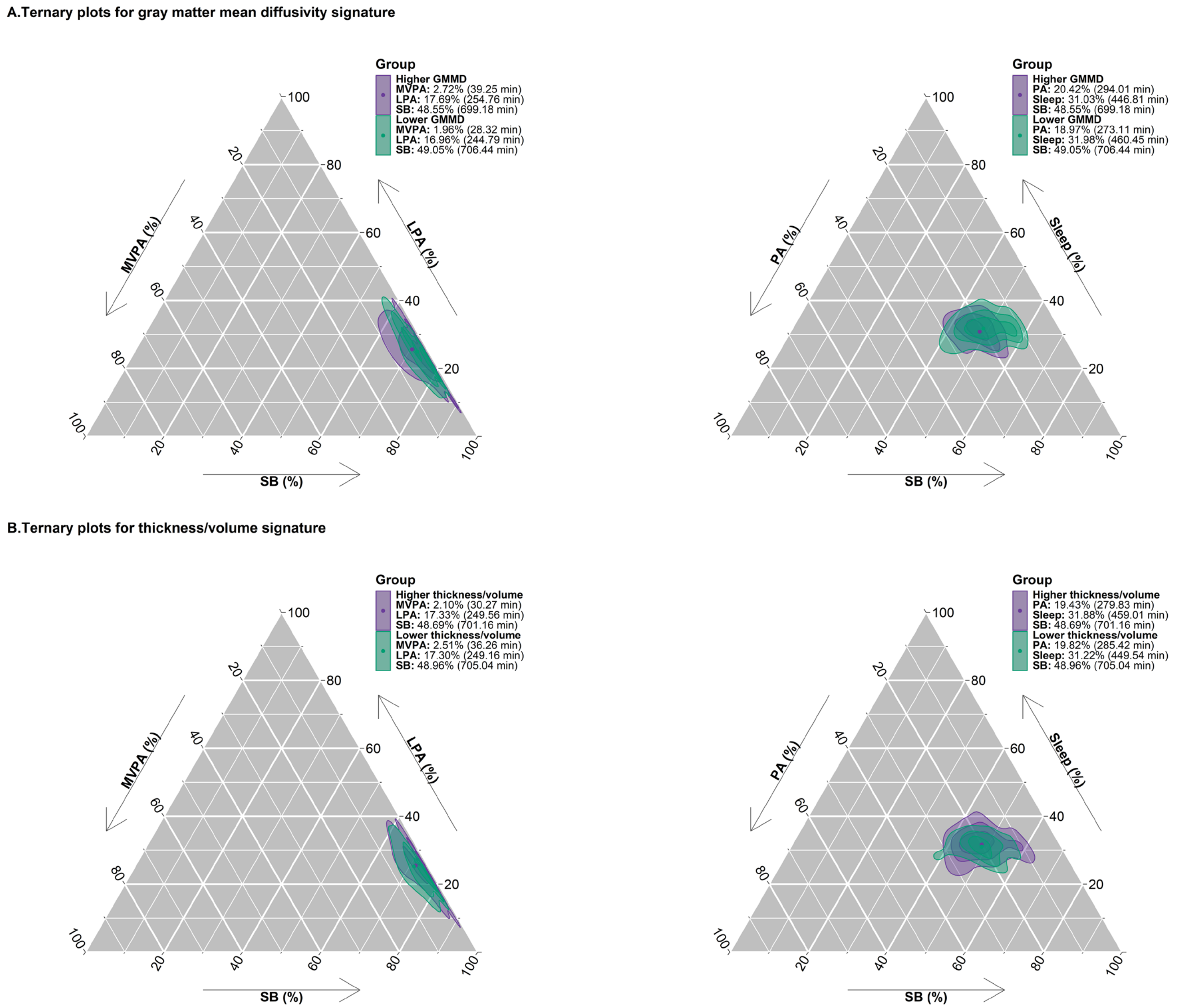

FIGURE 1 | Ternary plots for the daily movement behavior composition in MVPA, LPA, and SB (in %) and the daily movement behavior composition in PA, Sleep, SB (in %) stratified by higher and lower mean values of (A) gray matter mean diffusivity signature and (B) thickness/volume signature. Values represent compositional geometric means (expressed in % and minutes/day). The crosshair marks the compositional geometric mean in %. Concentric rings represent the 25%, 50%, and 75% estimated confidence regions for the data. LPA, light physical activity; MVPA, moderate-to-vigorous physical activity; PA, physical activity; SB, sedentary behavior.

day of MVPA and 249.16 min/day of LPA) compared with the higher thickness/volume group (19.43%, 279.83 min/day; 30.27 min/day of MVPA and 249.56 min/day of LPA), whereas proportions of time spent in SB were similar across groups.

3.2 | Associations of 24-h Movement Behaviors With Cognition

As shown in Figure 2, no overall associations were observed between 24-h movement behavior compositions and cognitive domains, including attentional/inhibitory control, episodic memory, processing speed, visuospatial processing, and working memory (all $p > 0.05$). Consistently, the ANOVA/F-tests for the direct effect of time-use composition were not statistically significant for any cognitive domain (all $p > 0.05$; Table S1).

Pearson correlations between cognitive outcomes and AD brain signatures are presented in Figure S1. No significant correlations were observed between cognitive domains and either GMMD or thickness/volume signatures (all $p > 0.05$). Correlation coefficients ranged from −0.06 to 0.13 for GMMD and from −0.09 to 0.04 for thickness/volume signatures.

3.3 | Moderation and Subgroup Analyses

We examined whether AD brain signatures moderated the associations between 24-h movement behavior composition and cognitive outcomes. ANOVA F-tests were first used to examine whether AD brain signatures moderated the association between the overall 24-h movement behavior composition and cognitive outcomes. As shown in Table S1, GMMD significantly

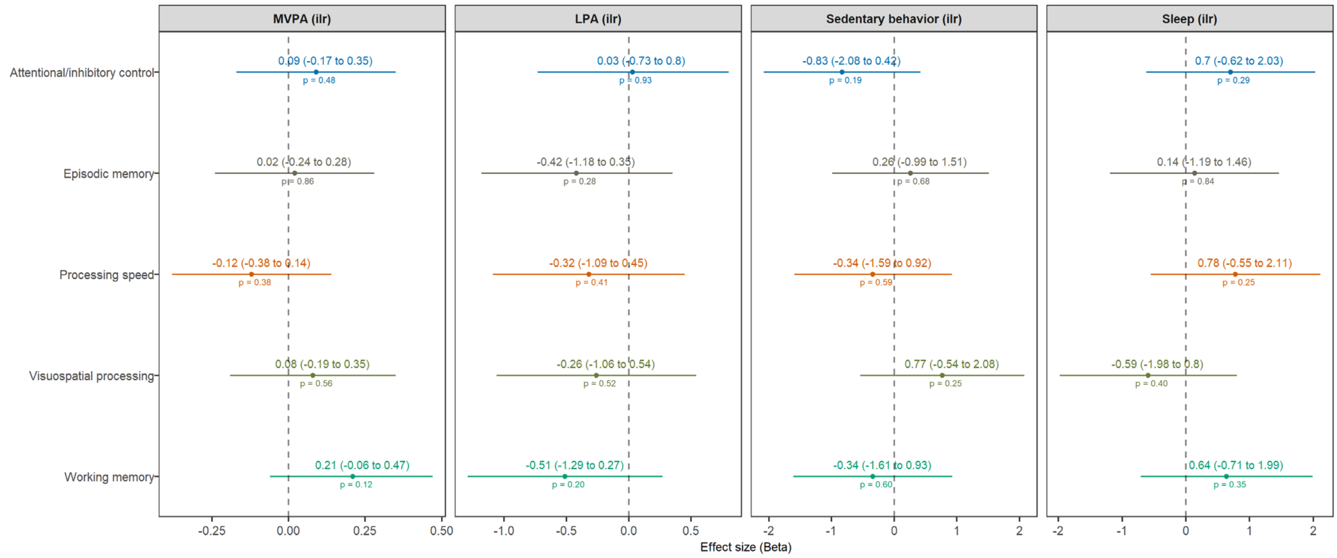

FIGURE 2 | Linear regression models showing the associations of movement behaviors (using a compositional analysis approach) with cognitive domains. For each behavior, independent models were constructed. Models for cognitive domains were adjusted for sex, age, and education level. In all models, the isometric log-ratio coordinates of the movement behaviors were included as predictor variables. The figure reports the results for the coordinate of the dominant behavior (listed in the header) relative to the other behaviors. B: Unstandardized beta coefficients indicate how much the outcome changes for a one-unit increase in the respective isometric log-ratio coordinate. This reflects a proportional increase in the dominant behavior relative to the rest of behaviors.

moderated the associations between the overall movement behavior composition and attentional/inhibitory control ($F = 2.89$, $p = 0.041$), episodic memory ($F = 5.16$, $p = 0.003$), processing speed ($F = 3.00$, $p = 0.036$), and working memory ($F = 4.43$, $p = 0.006$). In contrast, the thickness/volume signature did not significantly moderate the association between overall movement behavior composition and any cognitive domain (all $p > 0.05$). Behavior-specific pivot ILR interaction models are presented in Table S2. These models indicated that GMMD, but not the thickness/volume signature, moderated several behavior-specific associations between movement behavior composition and cognition.

As shown in Figure 3, specifically, behavior-specific pivot ILR models show how GMMD significantly moderated several associations between 24-h movement behaviors and cognitive outcomes. A consistent pattern of moderation was observed for episodic memory, for which significant interactions were found for MVPA (p for interaction $= 0.004$), LPA ($p = 0.008$), and sleep ($p = 0.009$), whereas no interaction was observed for SB ($p = 0.10$). Likewise, LPA showed the most consistent pattern of moderation across cognitive domains, with significant interactions for attention/inhibitory control ($p = 0.04$), episodic memory ($p = 0.008$), processing speed ($p = 0.04$), and working memory ($p = 0.02$), while no interaction was observed for visuospatial processing ($p = 0.70$). In contrast, no significant interactions were observed for SB across any cognitive domain (all p for interaction > 0.10), while sleep only moderated the association with episodic memory ($p = 0.009$).

Simple slope analyses of the significant interactions indicated that among participants with higher GMMD levels, reallocating time proportionally toward LPA was associated with poorer working memory ($B = -1.23$, 95% CI: -2.16 to -0.31, $p = 0.01$). Although the remaining simple slopes did not reach statistical significance, they consistently suggested that the direction of the associations between movement behaviors and cognition differed according to GMMD levels.

3.4 | Exploratory Analyses

Exploratory time-reallocation analyses were conducted following the significant GMMD moderation findings (Figure S2). Among individuals with lower GMMD levels, reallocating time proportionally toward LPA was significantly associated with better attentional/inhibitory control performance (ILR_LPA $B = 1.01$; $p = 0.05$). Similarly, one-to-one reallocation models showed that reallocating 30 min from SB to LPA was associated with 0.21 higher attentional/inhibitory control performance.

In contrast, among individuals with higher GMMD signature levels, reallocating time proportionally toward LPA was significantly associated with poorer working memory performance (ILR_LPA $B = -1.23$; $p = 0.01$). Consistent with this pattern, one-to-one reallocation models showed that reallocating 30 min from SB to LPA was associated with 0.12 lower working memory performance. Additional exploratory reallocation plots did not consistently reach statistical significance.

4 | Discussion

The present study investigated the associations between 24-h movement behaviors and multiple cognitive domains related to aging (i.e., attention/inhibitory control, episodic memory, processing speed, visuospatial processing and working memory) in cognitively unimpaired older adults, and whether these associations differed according to AD-related brain signatures reflecting brain reserve (i.e., GMMD and thickness/volume

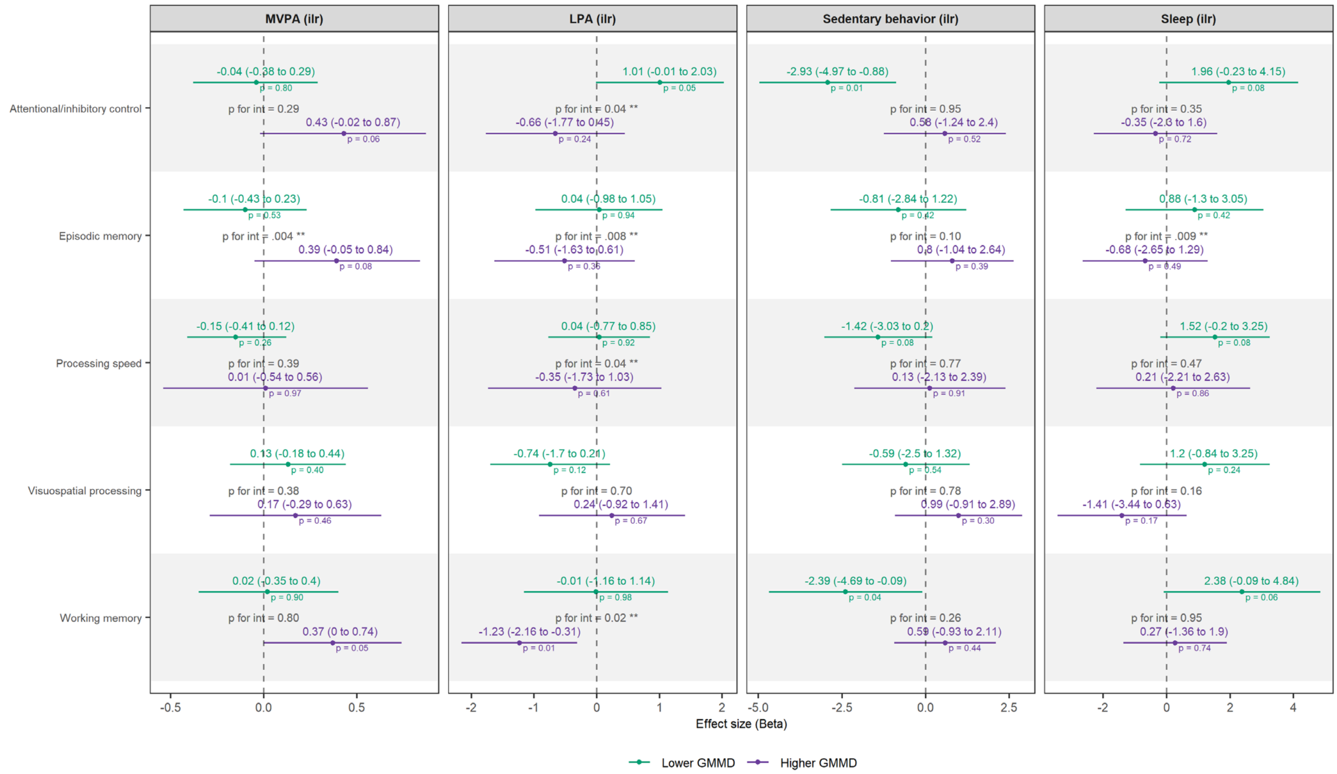

FIGURE 3 | Linear regression models showing the associations of movement behaviors (using a compositional analysis approach) with cognitive domains stratified by high and low values of gray matter mean diffusivity signature. Participants were classified as "high" or "low" based on the median of the total z-score signature level. For each behavior, independent models were constructed, adjusted for sex and education level, in which the isometric log-ratio coordinates of the movement behaviors were included as predictor variables. *p*-values for interaction were obtained from models including the continuous GMMD signature as moderator. The figure displays the results for the coordinate of the dominant behavior relative to the other behaviors. B: Unstandardized beta coefficients indicate how much the outcome changes for a one-unit increase in the respective isometric log-ratio coordinates. This reflects a proportional increase in the dominant behavior relative to the rest of behaviors.

signatures). While no overall associations were observed between 24-h movement behaviors and cognitive function, AD-related microstructural vulnerability, indexed by GMMD levels, emerged as a key modifier of these relationships. Moderation analyses were most consistently observed for episodic memory, for which three of the four movement behaviors showed significant interactions with GMMD. Likewise, LPA exhibited the broadest pattern of moderation across cognitive domains, with significant interactions for attention/inhibitory control, episodic memory, processing speed, and working memory. These findings suggest that differences in brain reserve microstructural properties may partly help explain the heterogeneous associations between 24-h movement behaviors and cognition in cognitively unimpaired older adults.

We did not find any significant associations between 24-h movement behaviors and cognitive outcomes, in contrast to some prior studies reporting positive associations between time-use composition and cognitive function [5–8, 19, 66] but in line with others reporting null findings [9, 20]. Importantly, the literature examining these relationships, particularly when considering the potential effects of reallocating time spent in one behavior for another, remains limited and methodologically heterogeneous, which may contribute to the inconsistency of findings among studies applying CoDA [12]. First, variations in sample characteristics are likely to play a role and differences in baseline levels of physical activity may be particularly relevant. For instance, some studies have reported lower levels of MVPA at baseline (e.g., 31 min/day [5]), whereas others reported higher levels (e.g., 89 min/day [9]). These differences may influence the magnitude of the observed associations, as reallocating time to MVPA in less active samples represents a larger proportional increase and may therefore have a stronger association with cognitive outcomes, whereas in more active samples, similar reallocations may reflect smaller relative changes and thus weaker or non-detectable associations [9]. Second, differences in cognitive assessment may further contribute to the inconsistency across studies. Prior research has often relied on single cognitive tests [20] or brief batteries [5, 9], whereas the present study employed a comprehensive neuropsychological assessment, allowing for the examination of multiple cognitive domains. This approach represents a methodological strength but at the same time, it may reveal that associations between movement behaviors and cognition are not uniform across domains, which could partly explain why studies using single measures sometimes report associations that are not consistently observed across broader cognitive profiles. Third, the absence of overall associations in the total sample may also reflect underlying heterogeneity in movement–cognition relationships according to levels of brain reserve. As suggested by our moderation analyses and illustrated in the reallocation plots, participants with higher and lower GMMD levels often showed associations in opposite directions. Therefore, when analyzed together as a single group, these subgroup-specific patterns may have attenuated or "washed out" overall associations, resulting in flat or non-significant average effects. Nevertheless, methodological differences such as sample size and statistical power may also influence the detection of associations. Studies with larger samples (e.g., $n = 384$ [9] or $n = 8.608$ [6]) may be better positioned to detect subtle movement–cognition relationships.

Beyond these methodological differences, one potential explanation for the inconsistencies is the assumption that the cognitive effects of movement behaviors do not consider underlying differences in brain characteristics. For example, Dumuid et al. (2022) observed larger significant associations between executive function and global cognition with MVPA in APOE ε4 carriers, highlighting the importance of accounting for underlying biological vulnerability when examining movement–cognition relationships [7]. It aligns with prior evidence suggesting that the composition of 24-h movement behaviors may play a particularly important role in preserving cognitive function among individuals at elevated risk for dementia [67]. According to levels of AD brain signatures, our findings indicate that the association between 24-h movement behaviors and cognition differs according to the degree of underlying microstructural integrity, indexed by GMMD levels, but not by thickness/volume signature. In individuals with better microstructural brain morphology (i.e., lower GMMD), lower-intensity activity (i.e., LPA) appeared sufficient to support certain cognitive functions, as observed for attentional/inhibitory control, episodic memory, processing speed and working memory. In contrast, in the context of greater AD-related microstructural vulnerability (i.e., higher GMMD), higher amount of high-intensity physical activity (i.e., MVPA) was associated with better episodic memory, interestingly, a domain vulnerable to early microstructural alterations [68] and a result in line with our initial hypothesis. These findings suggest that memory processes may be especially responsive to interactions between lifestyle behaviors and underlying brain properties [69]. Overall, these patterns support the notion that movement–cognition relationships are conditional upon the brain's capacity to respond to behavioral stimuli.

A notable finding of this study is that moderation effects were observed for microstructural brain signatures (GMMD), but not for macrostructural measures (thickness/volume). This may be explained by the greater sensitivity of diffusion-based metrics to early brain alterations, as microstructural changes often may occur earlier in the disease continuum than detectable changes in cortical thickness or regional volume [70]. For example, increases in mean diffusivity have been linked to early neurodegenerative processes, including amyloid-related changes, and may precede measurable brain atrophy [31]. In contrast, macrostructural measures may remain relatively preserved in cognitively unimpaired older adults, limiting their ability to capture subtle brain vulnerability [2]. Notably, recent evidence suggests that microstructural alterations are more closely associated with neurobehavioral performance than macrostructural measures, supporting their role as early and sensitive markers of brain changes relevant for cognitive function [71]. Together, these findings suggest that microstructural markers may better identify early stages of brain vulnerability during which responsiveness to lifestyle factors is still detectable.

This study possesses several strengths: (i) it used objective measurement of 24-h movement behaviors through accelerometry and applied CoDA analysis, allowing appropriate handling of co-dependent time-use data, (ii) comprehensive neuropsychological assessment, with multiple tests within each cognitive subdomain, (iii) the use of composite scores of structural and diffusion brain imaging to assess AD brain signatures and (iv) finally, the isotemporal reallocation plots reflect theoretical time substitutions within broadly overlapping movement distributions across AD signatures subgroups, demonstrating that even within shared time-use ranges, the predicted cognitive outcomes following time reallocation can differ according to microstructural brain integrity. However, several limitations must be considered when interpreting our findings. First, the cross-sectional design precludes establishing temporal precedence or causality. Bidirectional associations and reverse causality cannot be ruled out, as subclinical neurodegenerative processes or subtle cognitive changes could themselves alter daily PA patterns and sleep quality. Consequently, these results should be viewed as a cross-sectional proof-of-concept rather than an established prospective causal pathway. Second, the modest sample size in stratified subgroup analyses may have reduced statistical power, increasing the risk of Type II errors. Lastly, formal adjustments for multiple testing across cognitive outcomes were not performed. Given the exploratory nature of these moderation analyses, overly conservative corrections were avoided to prevent further inflating Type II error. We therefore prioritized the direction, magnitude, and precision of effect estimates and their confidence intervals, leaving these initial patterns to be confirmed in larger prospective cohorts.

In conclusion, although no overall associations were observed between 24-h movement behaviors and cognitive function, AD-related brain microstructural integrity, indexed by the GMMD signature, modified these associations, particularly for episodic memory and LPA. These findings highlight the importance of considering underlying brain integrity when investigating movement behavior–cognition associations and may help explain the heterogeneous findings reported in the literature.

5 | Perspective

Advancing toward personalized movement prescriptions in brain aging requires moving beyond cross-sectional observations toward prospective and experimental frameworks. Prospective longitudinal cohorts with repeated assessments of brain integrity, accelerometry-derived 24-h movement behaviors, and cognitive trajectories are essential to establish temporal precedence and rule out reverse causality. Furthermore, randomized controlled trials should investigate whether prescribing targeted daily time shifts (e.g., substituting sedentary time with PA) yields direct cognitive benefits in individuals with brain AD vulnerability. Extending these designs to larger, more clinically diverse cohorts will clarify whether intensity-specific recommendations differ across vulnerability profiles, such as older adults with subjective cognitive decline, mild cognitive impairment, or cardiometabolic risk factors. Finally, methodological priorities must focus on longitudinal compositional moderation frameworks, harmonized cognitive composite construction, and the integration of microstructural diffusion signatures with fluid biomarkers (e.g., plasma amyloid, tau, vascular, and inflammatory markers) to map the mechanistic pathways linking daily time use to cognitive function.

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Ethics Statement

The trial protocol followed the principles of the Declaration of Helsinki and was approved by the Research Ethics Board of the Andalusian Health Service (CEIM/CEI Provincial de Granada; #2317-N-19) on May 25, 2020. All participants provided written informed consent.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data of this study are available from the corresponding author, upon reasonable request. Details of the methodology and protocols of the AGUEDA project have been described elsewhere [35] and deposited at GitHub (https://github.com/aguedaprojectugr).

References

1. W. Y. W. Yau, D. R. Kirn, J. S. Rabin, et al., "Physical Activity as a Modifiable Risk Factor in Preclinical Alzheimer's Disease," *Nature Medicine* 31 (2025): 4075–4083, https://doi.org/10.1038/s41591-025-03955-6.

2. R. Tang, C. E. Franz, R. L. Hauger, et al., "Early Cortical Microstructural Changes in Aging Are Linked to Vulnerability to Alzheimer's Disease Pathology," *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging* 9, no. 10 (2024): 975–985, https://doi.org/10.1016/j.bpsc.2024.05.012.

3. Y. Stern, E. M. Arenaza-Urquijo, D. Bartrés-Faz, et al., "Whitepaper: Defining and Investigating Cognitive Reserve, Brain Reserve, and Brain Maintenance," *Alzheimer's & Dementia* 16 (2020): 1305–1311, https://doi.org/10.1016/j.jalz.2018.07.219.

4. M. Wirth, C. M. Haase, S. Villeneuve, J. Vogel, and W. J. Jagust, "Neuroprotective Pathways: Lifestyle Activity, Brain Pathology, and Cognition in Cognitively Normal Older Adults," *Neurobiology of Aging* 35, no. 8 (2014): 1873–1882, https://doi.org/10.1016/j.neurobiolaging.2014.02.015.

5. A. M. Collins, M. L. Mellow, A. E. Smith, et al., "24-Hour Time Use and Cognitive Performance in Late Adulthood: Results From the Investigating Gains in Neurocognition in an Intervention Trial of Exercise (IGNITE) Study," *Age and Ageing* 54, no. 4 (2025): 166–168, https://doi.org/10.1093/ageing/afaf072.

6. N. Feter, D. De Paula, R. C. P. Dos Reis, et al., "Association Between 24-Hour Movement Behavior and Cognitive Function in Brazilian Middle-Aged and Older Adults: Findings From the ELSA-Brasil," *Innovation in Aging* 7, no. 3 (2023): igad030, https://doi.org/10.1093/geroni/igad030.

7. D. Dumuid, M. L. Mellow, T. Olds, et al., "Does APOE ϵ4 Status Change How 24-Hour Time-Use Composition Is Associated With Cognitive Function? An Exploratory Analysis Among Middle-To-Older Adults," *Journal of Alzheimer's Disease* 88, no. 3 (2022): 1157–1165, https://doi.org/10.3233/JAD-220181.

8. P. J. Marent, G. Cardon, D. Dumuid, G. Albouy, and J. van Uffelen, "24-Hour Movement Behaviours Are Cross-Sectionally Associated With Cognitive Function in Healthy Adults Aged 55 Years and Older," *Scientific Reports* 15, no. 1 (2025): 38619, https://doi.org/10.1038/s41598-025-22541-x.

9. M. L. Mellow, D. Dumuid, A. T. Wade, et al., "Twenty-Four-Hour Time-Use Composition and Cognitive Function in Older Adults: Cross-Sectional Findings of the ACTIVate Study," *Frontiers in Human Neuroscience* 16 (2022): 1051793, https://doi.org/10.3389/fnhum.2022.1051793.

10. A. Sclafani, J. H. Migueles, M. Olvera-Rojas, et al., "Association of 24 h-Movement Behaviors With Cerebral and Hippocampal Amyloid Beta Levels and Executive Function in Cognitively Normal Older Adults: A Compositional Data Analysis From the AGUEDA Trial," *Journal of Science and Medicine in Sport* 28, no. 9 (2025): 748–756, https://doi.org/10.1016/j.jsams.2025.04.011.

11. G. M. Balbim, N. C. Boa Sorte Silva, R. S. Falck, A. F. Kramer, M. W. Voss, and T. Liu-Ambrose, "24-Hour Activity Cycle Behaviors and Gray Matter Volume in Mild Cognitive Impairment," *Alzheimer's & Dementia* 21, no. 7 (2025): e70496, https://doi.org/10.1002/alz.70496.

12. M. L. Mellow, D. Dumuid, T. Olds, et al., "Cross-Sectional Associations Between 24-Hour Time-Use Composition, Grey Matter Volume and Cognitive Function in Healthy Older Adults," *International Journal of Behavioral Nutrition and Physical Activity* 21, no. 1 (2024): 11, https://doi.org/10.1186/s12966-023-01557-4.

13. S. Sabia, A. Fayosse, J. Dumurgier, et al., "Association of Sleep Duration in Middle and Old Age With Incidence of Dementia," *Nature Communications* 12, no. 1 (2021): 2289, https://doi.org/10.1038/s41467-021-22354-2.

14. S. Yan, W. Fu, C. Wang, et al., "Association Between Sedentary Behavior and the Risk of Dementia: A Systematic Review and Meta-Analysis," *Translational Psychiatry* 10 (2020): 112, https://doi.org/10.1038/s41398-020-0799-5.

15. S. J. Blondell, R. Hammersley-Mather, and J. L. Veerman, "Does Physical Activity Prevent Cognitive Decline and Dementia?: A Systematic Review and Meta-Analysis of Longitudinal Studies," *BMC Public Health* 14 (2014): 510, https://doi.org/10.1186/1471-2458-14-510.

16. L. G. Johnson, M. L. Butson, R. C. Polman, et al., "Light Physical Activity Is Positively Associated With Cognitive Performance in Older Community Dwelling Adults," *Journal of Science and Medicine in Sport* 19, no. 11 (2016): 877–882, https://doi.org/10.1016/j.jsams.2016.02.002.

17. S. Halloway, K. Arfanakis, J. E. Wilbur, M. E. Schoeny, and S. J. Pressler, "Accelerometer Physical Activity Is Associated With Greater Gray Matter Volumes in Older Adults Without Dementia or Mild Cognitive Impairment," *Journals of Gerontology. Series B, Psychological*

Sciences and Social Sciences 74, no. 7 (2019): 1142–1151, https://doi.org/10.1093/geronb/gby010.

18. L. Coelho, K. Hauck, K. McKenzie, et al., "The Association Between Sedentary Behavior and Cognitive Ability in Older Adults," *Aging Clinical and Experimental Research* 32, no. 11 (2020): 2339–2347, https://doi.org/10.1007/s40520-019-01460-8.

19. J. Wei, R. Hou, L. Xie, et al., "Sleep, Sedentary Activity, Physical Activity, and Cognitive Function Among Older Adults: The National Health and Nutrition Examination Survey, 2011–2014," *Journal of Science and Medicine in Sport* 24, no. 2 (2021): 189–194, https://doi.org/10.1016/j.jsams.2020.09.013.

20. Y. Wu, D. E. Rosenberg, M. A. Greenwood-Hickman, et al., "Analysis of the 24-h Activity Cycle: An Illustration Examining the Association With Cognitive Function in the Adult Changes in Thought Study," *Frontiers in Psychology* 14 (2023): 1083344, https://doi.org/10.3389/fpsyg.2023.1083344.

21. A. Miatke, T. Olds, C. Maher, et al., "The Association Between Reallocations of Time and Health Using Compositional Data Analysis: A Systematic Scoping Review With an Interactive Data Exploration Interface," *International Journal of Behavioral Nutrition and Physical Activity* 20 (2023): 127, https://doi.org/10.1186/s12966-023-01526-x.

22. C. M. Stillman, I. Esteban-Cornejo, B. Brown, C. M. Bender, and K. I. Erickson, "Effects of Exercise on Brain and Cognition Across Age Groups and Health States," *Trends in Neurosciences* 43, no. 7 (2020): 533–543, https://doi.org/10.1016/j.tins.2020.04.010.

23. Y. Stern, C. A. Barnes, C. Grady, R. N. Jones, and N. Raz, "Brain Reserve, Cognitive Reserve, Compensation, and Maintenance: Operationalization, Validity, and Mechanisms of Cognitive Resilience," *Neurobiology of Aging* 83 (2019): 124–129, https://doi.org/10.1016/j.neurobiolaging.2019.03.022.

24. P. Vemuri, T. G. Lesnick, D. S. Knopman, et al., "Amyloid, Vascular, and Resilience Pathways Associated With Cognitive Aging," *Annals of Neurology* 86, no. 6 (2019): 866–877, https://doi.org/10.1002/ana.25600.

25. M. E. Williams, J. A. Elman, L. K. Mcevoy, et al., "12-Year Prediction of Mild Cognitive Impairment Aided by Alzheimer's Brain Signatures at Mean Age 56," *Brain Communications* 3, no. 3 (2021): fcab167, https://doi.org/10.1093/braincomms/fcab167.

26. A. M. Racine, M. Brickhouse, D. A. Wolk, and B. C. Dickerson, "The Personalized Alzheimer's Disease Cortical Thickness Index Predicts Likely Pathology and Clinical Progression in Mild Cognitive Impairment," *Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring* 10 (2018): 301–310, https://doi.org/10.1016/j.dadm.2018.02.007.

27. B. C. Dickerson, T. R. Stoub, R. C. Shah, et al., "Alzheimer-Signature MRI Biomarker Predicts AD Dementia in Cognitively Normal Adults," 2011, www.neurology.org.

28. J. Hwang, C. M. Kim, S. Jeon, et al., "Prediction of Alzheimer's Disease Pathophysiology Based on Cortical Thickness Patterns," *Alzheimer's & Dementia (Amsterdam, Netherlands)* 2 (2016): 58–67, https://doi.org/10.1016/j.dadm.2015.11.008.

29. M. R. Sabuncu, R. S. Desikan, J. Sepulcre, et al., "The Dynamics of Cortical and Hippocampal Atrophy in Alzheimer Disease," *Archives of Neurology* 68, no. 8 (2011): 1040–1048, https://doi.org/10.1001/archneurol.2011.167.

30. D. K. Overson, T. K. Truong, J. R. Petrella, et al., "Predicting Risk of Impending Cognitive Decline in Asymptomatic Individuals With Early Alzheimer's Disease: Insights From Cortical Diffusion MRI," *Imaging Neuroscience* 3 (2025): IMAG.a.1037, https://doi.org/10.1162/imag.a.1037.

31. J. Debatisse, F. Leng, A. Ashraf, and P. Edison, "Cortical Diffusivity, a Biomarker for Early Neuronal Damage, Is Associated With Amyloid-β Deposition: A Pilot Study," *Cells* 14, no. 3 (2025): 155, https://doi.org/10.3390/cells14030155.

32. K. B. Casaletto, M. A. Rentería, J. Pa, et al., "Late-Life Physical and Cognitive Activities Independently Contribute to Brain and Cognitive Resilience," *Journal of Alzheimer's Disease* 74, no. 1 (2020): 363–376, https://doi.org/10.3233/JAD-191114.

33. Q. Tian, J. A. Schrack, B. A. Landman, S. M. Resnick, and L. Ferrucci, "Longitudinal Associations of Absolute Versus Relative Moderate-To-Vigorous Physical Activity With Brain Microstructural Decline in Aging," *Neurobiology of Aging* 116 (2022): 25–31, https://doi.org/10.1016/j.neurobiolaging.2022.04.007.

34. D. Dumuid, Ž. Pedišić, J. Palarea-Albaladejo, J. A. Martín-Fernández, K. Hron, and T. Olds, "Compositional Data Analysis in Time-Use Epidemiology: What, Why, How," *International Journal of Environmental Research and Public Health* 17, no. 7 (2020): 2220, https://doi.org/10.3390/ijerph17072220.

35. P. Solis-Urra, C. Molina-Hidalgo, Y. García-Rivero, et al., "Active Gains in Brain Using Exercise During Aging (AGUEDA): Protocol for a Randomized Controlled Trial," *Frontiers in Human Neuroscience* 17 (2023): 1168549, https://doi.org/10.3389/fnhum.2023.1168549.

36. K. Tomioka, J. Iwamoto, K. Saeki, and N. Okamoto, "Reliability and Validity of the International Physical Activity Questionnaire (IPAQ) in Elderly Adults: The Fujiwara-Kyo Study," *Journal of Epidemiology* 21, no. 6 (2011): 459–465, https://doi.org/10.2188/jea.JE20110003.

37. M. Muñoz-García, S. Cervantes, C. Razquin, et al., "Validation Study of a Spanish Version of the Modified Telephone Interview for Cognitive Status (STICS-m)," *Gaceta Sanitaria* 33, no. 5 (2019): 415–420, https://doi.org/10.1016/j.gaceta.2018.05.004.

38. S. R. de Beaman, P. E. Beaman, C. Garcia-Peña, et al., "Validation of a Modified Version of the Mini-Mental State Examination (MMSE) in Spanish," *Aging, Neuropsychology, and Cognition* 11, no. 1 (2004): 1–11, https://doi.org/10.1076/anec.11.1.1.29366.

39. N. Ojeda, R. del Pino, N. Ibarretxe-Bilbao, D. J. Schretlen, and J. Pena, "Montreal Cognitive Assessment Test: Normalization and Standardization for Spanish Population," *Revista de Neurologia* 63, no. 11 (2016): 488–496.

40. M. I. Fernández-San Martín, C. Andrade, J. Molina, et al., "Validation of the Spanish Version of the Geriatric Depression Scale (GDS) in Primary Care," *International Journal of Geriatric Psychiatry* 17, no. 3 (2002): 279–287, https://doi.org/10.1002/gps.588.

41. V. T. Van Hees, Z. Fang, J. Langford, et al., "Autocalibration of Accelerometer Data for Free-Living Physical Activity Assessment Using Local Gravity and Temperature: An Evaluation on Four Continents," *Journal of Applied Physiology (Bethesda, MD: 1985)* 117, no. 7 (2014): 738–744, https://doi.org/10.1152/JAPPLPHYSIOL.00421.2014.

42. V. T. van Hees, F. Renström, A. Wright, et al., "Estimation of Daily Energy Expenditure in Pregnant and Non-Pregnant Women Using a Wrist-Worn Tri-Axial Accelerometer," *PLoS One* 6, no. 7 (2011): e22922, https://doi.org/10.1371/JOURNAL.PONE.0022922.

43. M. Hildebrand, B. H. Hansen, V. T. van Hees, and U. Ekelund, "Evaluation of Raw Acceleration Sedentary Thresholds in Children and Adults," *Scandinavian Journal of Medicine & Science in Sports* 27, no. 12 (2017): 1814–1823, https://doi.org/10.1111/SMS.12795.

44. M. Hildebrand, V. T. Van Hees, B. H. Hansen, and U. Ekelund, "Age Group Comparability of Raw Accelerometer Output From Wrist- and Hip-Worn Monitors," *Medicine and Science in Sports and Exercise* 46, no. 9 (2014): 1816–1824, https://doi.org/10.1249/MSS.0000000000000289.

45. J. H. Migueles, A. V. Rowlands, F. Huber, S. Sabia, and V. T. Van Hees, "GGIR: A Research Community–Driven Open Source R Package for Generating Physical Activity and Sleep Outcomes From Multi-Day Raw Accelerometer Data," *Journal of Measurement in Physical Behavior* 2, no. 3 (2019): 188–196, https://doi.org/10.1123/jmpb.2018-0063.

46. L. E. Oberlin, L. Wan, C. Kang, et al., "Cardiorespiratory Fitness Is Associated With Cognitive Function in Late Adulthood: Baseline

Findings From the IGNITE Study," *British Journal of Sports Medicine* 59 (2024): 167–176, https://doi.org/10.1136/bjsports-2024-108257.

47. R. M. Reitan, "Validity of the Trail Making Test as an Indicator of Organic Brain Damage," *Perceptual and Motor Skills* 8, no. 3 (1958): 271–276, https://doi.org/10.2466/pms.1958.8.3.271.

48. P. D. Zelazo, J. E. Anderson, J. Richler, K. Wallner-Allen, J. L. Beaumont, and S. Weintraub, "NIH Toolbox Cognition Battery (CB): Measuring Executive Function and Attention," *Monographs of the Society for Research in Child Development* 78, no. 4 (2013): 16–33, https://doi.org/10.1111/mono.12032.

49. Z. S. Nasreddine, N. A. Phillips, V. Bédirian, et al., "The Montreal Cognitive Assessment, MoCA: A Brief Screening Tool for Mild Cognitive Impairment," *Journal of the American Geriatrics Society* 53, no. 4 (2005): 695–699, https://doi.org/10.1111/j.1532-5415.2005.53221.x.

50. V. Alviarez-Schulze, G. Cattaneo, C. Pachón-García, et al., "Validation and Normative Data of the Spanish Version of the Rey Auditory Verbal Learning Test and Associated Long-Term Forgetting Measures in Middle-Aged Adults," *Frontiers in Aging Neuroscience* 14 (2022): 809019, https://doi.org/10.3389/fnagi.2022.809019.

51. X. Zhang, L. Lv, G. Min, Q. Wang, Y. Zhao, and Y. Li, "Overview of the Complex Figure Test and Its Clinical Application in Neuropsychiatric Disorders, Including Copying and Recall," *Frontiers in Neurology* 12 (2021): 680474, https://doi.org/10.3389/fneur.2021.680474.

52. D. Wechsler, "Wechsler Adult Intelligence Scale—Fourth Edition Administration and Scoring Manual," 2008, https://doi.org/10.1177/073428298300100310.

53. T. A. Girard, B. N. Axelrod, R. Patel, and J. R. Crawford, "Wechsler Adult Intelligence Scale–IV Dyads for Estimating Global Intelligence," *Assessment* 22, no. 4 (2015): 441–448, https://doi.org/10.1177/1073191114551551.

54. K. I. Erickson, M. W. Voss, R. S. Prakash, et al., "Exercise Training Increases Size of Hippocampus and Improves Memory," *Proceedings of the National Academy of Sciences of the United States of America* 108, no. 7 (2011): 3017–3022, https://doi.org/10.1073/pnas.1015950108.

55. D. S. Tulsky, N. Carlozzi, N. D. Chiaravalloti, et al., "NIH Toolbox Cognition Battery (NIHTB-CB): List Sorting Test to Measure Working Memory," *Journal of the International Neuropsychological Society* 20, no. 6 (2014): 599–610, https://doi.org/10.1017/S135561771400040X.

56. A. M. Owen, K. M. McMillan, A. R. Laird, and E. Bullmore, "N-Back Working Memory Paradigm: A Meta-Analysis of Normative Functional Neuroimaging Studies," *Human Brain Mapping* (2005): 46–59, https://doi.org/10.1002/hbm.20131.

57. B. Fernandez-Gamez, P. Solis-Urra, A. Coca-Pulido, et al., "Effect of a 24-Week Resistance Exercise Intervention on Cognitive Function in Cognitively Normal Older Adults: The AGUEDA Randomized Controlled Trial," *Alzheimer's & Dementia* 22, no. 1 (2026): e71019, https://doi.org/10.1002/alz.71019.

58. L. K. McEvoy, C. Fennema-Notestine, J. C. Roddey, et al., "Alzheimer Disease: Quantitative Structural Neuroimaging for Detection and Prediction of Clinical and Structural Changes in Mild Cognitive Impairment," *Radiology* 251, no. 1 (2009): 195–205, https://doi.org/10.1148/radiol.2511080924.

59. E. Cavedo, S. Galluzzi, M. Pievani, M. Boccardi, and G. B. Frisoni, "Norms for Imaging Markers of Brain Reserve," *Journal of Alzheimer's Disease* 31, no. 3 (2012): 623–633, https://doi.org/10.3233/JAD-2012-111817.

60. M. E. Williams, J. A. Elman, T. R. Bell, et al., "Higher Cortical Thickness/Volume in Alzheimer's-Related Regions: Protective Factor or Risk Factor?," *Alzheimer's & Dementia* 19, no. S10 (2023): 185–194, https://doi.org/10.1002/alz.081784.

61. J. Sanchez-Martinez, P. Solis-Urra, B. Fernandez-Gamez, et al., "Effects of a 24-Week Resistance Exercise Program on Alzheimer's Disease Brain Signatures in Cognitively Unimpaired Older Adults: A Secondary Analysis of the AGUEDA Randomized Controlled Trial," *Age and Ageing* 55, no. 4 (2026): afag086, https://doi.org/10.1093/ageing/afag086.

62. K. Van den Boogaart and D. R. R. Tolosana, "Compositions: Compositional Data Analysis. R package version 2.0-1," 2021, http://www.stat.boogaart.de/compositions/.

63. M. Templ, K. Hron, and P. Filzmoser, "robCompositions: An R-Package for Robust Statistical Analysis of Compositional Data," 2011.

64. S. F. M. Chastin, J. Palarea-Albaladejo, M. L. Dontje, and D. A. Skelton, "Combined Effects of Time Spent in Physical Activity, Sedentary Behaviors and Sleep on Obesity and Cardio-Metabolic Health Markers: A Novel Compositional Data Analysis Approach," *PLoS One* 10, no. 10 (2015): e0139984, https://doi.org/10.1371/JOURNAL.PONE.0139984.

65. D. Dumuid, T. E. Stanford, J. A. Martin-Fernández, et al., "Compositional Data Analysis for Physical Activity, Sedentary Time and Sleep Research," *Statistical Methods in Medical Research* 27, no. 12 (2018): 3726–3738, https://doi.org/10.1177/0962280217710835.

66. J. Fanning, G. Porter, E. A. Awick, et al., "Replacing Sedentary Time With Sleep, Light, or Moderate-To-Vigorous Physical Activity: Effects on Self-Regulation and Executive Functioning," *Journal of Behavioral Medicine* 40, no. 2 (2017): 332–342, https://doi.org/10.1007/s10865-016-9788-9.

67. J. Carson Smith, K. A. Nielson, J. L. Woodard, M. Seidenberg, and S. M. Rao, "Physical Activity and Brain Function in Older Adults at Increased Risk for Alzheimer's Disease," *Brain Sciences* 3 (2013): 54–83, https://doi.org/10.3390/brainsci3010054.

68. L. Nyberg, M. Lövdén, K. Riklund, U. Lindenberger, and L. Bäckman, "Memory Aging and Brain Maintenance," *Trends in Cognitive Sciences* 16 (2012): 292–305, https://doi.org/10.1016/j.tics.2012.04.005.

69. T. Eisenstein, N. Giladi, T. Hendler, O. Havakuk, and Y. Lerner, "Hippocampal and Non-Hippocampal Correlates of Physically Active Lifestyle and Their Relation to Episodic Memory in Older Adults," *Neurobiology of Aging* 109 (2022): 100–112, https://doi.org/10.1016/j.neurobiolaging.2021.08.017.

70. F. Ji, O. Pasternak, K. K. Ng, et al., "White Matter Microstructural Abnormalities and Default Network Degeneration Are Associated With Early Memory Deficit in Alzheimer's Disease Continuum," *Scientific Reports* 9, no. 1 (2019): 4749, https://doi.org/10.1038/s41598-019-41363-2.

71. K. Singh, S. Barsoum, K. G. Schilling, Y. An, L. Ferrucci, and D. Benjamini, "Neuronal Microstructural Changes in the Human Brain Are Associated With Neurocognitive Aging," 2024, https://doi.org/10.1101/2024.01.11.575206.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** ANOVA *F*-tests for the association between 24-h movement behaviors and cognitive performance moderated by AD brain signatures. **Table S2:** Linear regression models showing the associations of movement behaviors with cognitive domains moderated by AD brain signatures. **Figure S1:** Pearson correlations between cognitive outcomes and AD signatures. Color bar represents the strength of the correlation as *r* values. The darker blue the color, the stronger the positive correlation between variables, and the darker red the color, the stronger the negative correlation between variables. Higher values of the gray matter mean diffusivity (GMMD) signature indicate greater microstructural brain disruption (worse brain reserve), whereas higher cortical thickness/volume signature values indicate better structural integrity. **Figure S2:** Pairwise isotemporal reallocations of time between movement behaviors in relation to cognitive outcomes showing significant gray matter mean diffusivity (GMMD) moderation. The figure illustrates the estimated differences in cognitive outcomes associated with reallocating time (minutes/day) between movement behaviors—Moderate-to-Vigorous Physical Activity (MVPA), Light Physical Activity (LPA), Sedentary Behavior (SB), and Sleep, among individuals with lower and higher GMMD signature levels. Only cognitive domains with significant interaction are displayed.